

3.8 Implicit Differentiation Worksheet

1. Find the slope of the line tangent to the graph of the ellipse $\frac{x^2}{2} + \frac{y^2}{36} = 1$ at the point $\left(\frac{3}{2}, 3\right)$.

2. Find the equation of the line tangent to the curve $5y^4 - x^3 = x^2y - 7$ at the point $(2, 1)$.

3. Compute $\frac{dy}{dx}$ for each of the following:

a) $xy^3 + x^5y = 42$

b) $\sin(x^2y^3) = 3x + 2y$

4. Find the slope of the line tangent to the curve $x^3 + y^3 = xy^2 + 5$ at the point $(1, 2)$.

5. Compute $\frac{dw}{dq}$ given that $w^2 q^3 = \sin(w^2) + q^5$.

6. If $xy = 1$ then find $\frac{dy}{dx}$.

7. Find $\frac{dy}{dx}$ if $5y^2 + \sin(y) = x^2$.

8. Find y'' if $4x^2 - 2y^2 = 9$.

9. Consider the curve given by $x^3 + y^3 = 6xy$.

a) Find $\frac{dy}{dx}$.

b) At what point in the first quadrant is the tangent line horizontal?